Analysis report

Holzinger-Swineford · MLR · listwise · N = 156 + 145 · lavaan 0.7.3

Table 1: Parameter estimates

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
Pasteur	visual	=~	x1		1.000	0.000	—	—	1.000	1.000	0.887
Pasteur	visual	=~	x2		0.394	0.197	1.996	.046	0.007	0.780	0.336
Pasteur	visual	=~	x3		0.570	0.201	2.838	.005	0.176	0.964	0.515
Pasteur	textual	=~	x4		1.000	0.000	—	—	1.000	1.000	0.823
Pasteur	textual	=~	x5		1.183	0.106	11.213	< .001	0.977	1.390	0.856
Pasteur	textual	=~	x6		0.875	0.093	9.388	< .001	0.692	1.058	0.838
Pasteur	speed	=~	x7		1.000	0.000	—	—	1.000	1.000	0.547
Pasteur	speed	=~	x8		1.125	0.262	4.298	< .001	0.612	1.638	0.682
Pasteur	speed	=~	x9		0.922	0.297	3.099	.002	0.339	1.505	0.551
Pasteur	x1	~~	x1		0.298	0.338	0.882	.378	-0.365	0.961	0.214
Pasteur	x2	~~	x2		1.334	0.179	7.441	< .001	0.982	1.685	0.887
Pasteur	x3	~~	x3		0.989	0.154	6.444	< .001	0.689	1.290	0.735
Pasteur	x4	~~	x4		0.425	0.073	5.867	< .001	0.283	0.567	0.322
Pasteur	x5	~~	x5		0.456	0.086	5.304	< .001	0.287	0.624	0.267
Pasteur	x6	~~	x6		0.290	0.055	5.267	< .001	0.182	0.398	0.297
Pasteur	x7	~~	x7		0.820	0.132	6.211	< .001	0.561	1.079	0.701
Pasteur	x8	~~	x8		0.510	0.101	5.048	< .001	0.312	0.708	0.535
Pasteur	x9	~~	x9		0.680	0.126	5.406	< .001	0.434	0.927	0.696
Pasteur	visual	~~	visual		1.097	0.378	2.899	.004	0.355	1.839	1.000
Pasteur	textual	~~	textual		0.894	0.160	5.577	< .001	0.580	1.209	1.000
Pasteur	speed	~~	speed		0.350	0.140	2.502	.012	0.076	0.623	1.000
Pasteur	visual	~~	textual		0.479	0.144	3.334	< .001	0.197	0.761	0.484
Pasteur	visual	~~	speed		0.185	0.076	2.449	.014	0.037	0.333	0.299
Pasteur	textual	~~	speed		0.182	0.076	2.402	.016	0.033	0.330	0.325
Pasteur	x1	~1			4.941	0.095	52.249	< .001	4.756	5.127	4.183
Pasteur	x2	~1			5.984	0.098	60.949	< .001	5.792	6.176	4.880
Pasteur	x3	~1			2.487	0.093	26.778	< .001	2.305	2.669	2.144
Pasteur	x4	~1			2.823	0.092	30.689	< .001	2.642	3.003	2.457
Pasteur	x5	~1			3.995	0.105	38.183	< .001	3.790	4.200	3.057
Pasteur	x6	~1			1.922	0.079	24.321	< .001	1.767	2.077	1.947
Pasteur	x7	~1			4.432	0.087	51.181	< .001	4.263	4.602	4.098
Pasteur	x8	~1			5.563	0.078	71.214	< .001	5.410	5.716	5.702
Pasteur	x9	~1			5.418	0.079	68.440	< .001	5.263	5.573	5.480
Pasteur	visual	~1			0.000	0.000	—	—	0.000	0.000	0.000
Pasteur	textual	~1			0.000	0.000	—	—	0.000	0.000	0.000

Table 1 (Parameter estimates)

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
Pasteur	speed	~1			0.000	0.000	—	—	0.000	0.000	0.000
Grant-White	visual	=~	x1		1.000	0.000	—	—	1.000	1.000	0.677
Grant-White	visual	=~	x2		0.736	0.190	3.870	< .001	0.363	1.109	0.517
Grant-White	visual	=~	x3		0.925	0.216	4.281	< .001	0.501	1.348	0.694
Grant-White	textual	=~	x4		1.000	0.000	—	—	1.000	1.000	0.866
Grant-White	textual	=~	x5		0.990	0.089	11.179	< .001	0.816	1.163	0.829
Grant-White	textual	=~	x6		0.963	0.091	10.588	< .001	0.785	1.142	0.826
Grant-White	speed	=~	x7		1.000	0.000	—	—	1.000	1.000	0.659
Grant-White	speed	=~	x8		1.226	0.162	7.560	< .001	0.908	1.544	0.796
Grant-White	speed	=~	x9		1.058	0.250	4.227	< .001	0.567	1.548	0.701
Grant-White	x1	~~	x1		0.715	0.182	3.938	< .001	0.359	1.071	0.542
Grant-White	x2	~~	x2		0.899	0.142	6.338	< .001	0.621	1.177	0.733
Grant-White	x3	~~	x3		0.557	0.122	4.568	< .001	0.318	0.796	0.519
Grant-White	x4	~~	x4		0.315	0.066	4.764	< .001	0.186	0.445	0.251
Grant-White	x5	~~	x5		0.419	0.071	5.876	< .001	0.279	0.559	0.312
Grant-White	x6	~~	x6		0.406	0.076	5.363	< .001	0.258	0.554	0.317
Grant-White	x7	~~	x7		0.600	0.100	5.995	< .001	0.404	0.797	0.566
Grant-White	x8	~~	x8		0.401	0.159	2.516	.012	0.089	0.714	0.367
Grant-White	x9	~~	x9		0.535	0.140	3.814	< .001	0.260	0.810	0.509
Grant-White	visual	~~	visual		0.604	0.194	3.105	.002	0.223	0.985	1.000
Grant-White	textual	~~	textual		0.942	0.162	5.807	< .001	0.624	1.260	1.000
Grant-White	speed	~~	speed		0.461	0.120	3.839	< .001	0.226	0.697	1.000
Grant-White	visual	~~	textual		0.408	0.115	3.535	< .001	0.182	0.634	0.541
Grant-White	visual	~~	speed		0.276	0.101	2.731	.006	0.078	0.474	0.523
Grant-White	textual	~~	speed		0.222	0.105	2.115	.034	0.016	0.427	0.336
Grant-White	x1	~1			4.930	0.095	51.696	< .001	4.743	5.117	4.293
Grant-White	x2	~1			6.200	0.092	67.416	< .001	6.020	6.380	5.599
Grant-White	x3	~1			1.996	0.086	23.195	< .001	1.827	2.164	1.926
Grant-White	x4	~1			3.317	0.093	35.625	< .001	3.135	3.500	2.959
Grant-White	x5	~1			4.712	0.096	48.986	< .001	4.524	4.901	4.068
Grant-White	x6	~1			2.469	0.094	26.277	< .001	2.285	2.653	2.182
Grant-White	x7	~1			3.921	0.086	45.819	< .001	3.753	4.089	3.805
Grant-White	x8	~1			5.488	0.087	63.174	< .001	5.318	5.659	5.246
Grant-White	x9	~1			5.327	0.085	62.571	< .001	5.160	5.494	5.196
Grant-White	visual	~1			0.000	0.000	—	—	0.000	0.000	0.000
Grant-White	textual	~1			0.000	0.000	—	—	0.000	0.000	0.000

Table 1 (Parameter estimates)

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
Grant-White	speed	~1			0.000	0.000	—	—	0.000	0.000	0.000

Note. Estimates and intervals are unstandardised; Std. all is fully standardised. Values are rounded only for display. A dash indicates unavailable output. 95% CI.

Table 2: Fit measures

Measure	Value
χ^2	115.851
df	48.000
p (χ^2)	< .001
CFI	0.923
TLI	0.885
RMSEA	0.097
RMSEA 90% CI lower	0.075
RMSEA 90% CI upper	0.120
SRMR	0.068
Scaled χ^2	121.741
Scaled df	48.000
p (scaled χ^2)	< .001
Robust CFI	0.921
Robust TLI	0.881
Robust RMSEA	0.099
Robust RMSEA 90% CI lower	0.077
Robust RMSEA 90% CI upper	0.120
Scaled CFI	0.914
Scaled TLI	0.872
Scaled RMSEA	0.101
Scaled RMSEA 90% CI lower	0.078
Scaled RMSEA 90% CI upper	0.124

Note. Scaled and robust statistics are labelled explicitly. Compare the appropriate version for your estimator.

Table 3: Explained variance (R^2)

Group	Left	R^2
Pasteur	x1	0.786
Pasteur	x2	0.113
Pasteur	x3	0.265
Pasteur	x4	0.678
Pasteur	x5	0.733
Pasteur	x6	0.703
Pasteur	x7	0.299
Pasteur	x8	0.465
Pasteur	x9	0.304
Grant-White	x1	0.458
Grant-White	x2	0.267
Grant-White	x3	0.481
Grant-White	x4	0.749
Grant-White	x5	0.688
Grant-White	x6	0.683
Grant-White	x7	0.434
Grant-White	x8	0.633
Grant-White	x9	0.491

Table 4: Measurement invariance

Stage	χ^2	df	p	CFI	RMSEA	SRMR	ΔCFI	ΔRMSEA	$\Delta\chi^2$	Δdf	Difference p	CFI variant	RMSEA variant
Configural	121.741	48	< .001	0.921	0.099	0.068	—	—	—	—	—	cfi.robust	rmsea.robust
Metric	125.997	54	< .001	0.920	0.093	0.072	-0.001	-0.005	6.567	6	.363	cfi.robust	rmsea.robust
Scalar	166.748	60	< .001	0.881	0.108	0.082	-0.039	0.014	40.838	6	< .001	cfi.robust	rmsea.robust
Strict	181.429	69	< .001	0.873	0.104	0.088	-0.008	-0.004	15.694	9	.074	cfi.robust	rmsea.robust

Note. Each stage is compared with the previous stage using lavaan's difference test. ΔCFI and ΔRMSEA are signed changes; the exact fit-index variants are reported. These statistics do not automatically establish invariance. Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001") Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001") Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")

Table 5: Reliability & validity

Group	Factor	Composite ω	AVE
Pasteur	visual	0.584	0.382
Pasteur	textual	0.877	0.707
Pasteur	speed	0.620	0.351
Grant-White	visual	0.661	0.400
Grant-White	textual	0.878	0.706
Grant-White	speed	0.766	0.521

Note. Unit-weight omega-type composite reliability and AVE use observed variances (semTools). HTMT2 uses absolute correlations and geometric means; ordinal indicators use polychoric correlations. Reverse-code items in the source data when appropriate. semTools 0.5.10.

Table 6: Discriminant validity

Group	Factor 1	Factor 2	HTMT2	Latent r	$\sqrt{\text{AVE}_1}$	$\sqrt{\text{AVE}_2}$	Fornell–Larcker
Pasteur	visual	textual	0.313	0.484	0.618	0.841	Yes
Pasteur	visual	speed	0.292	0.299	0.618	0.593	Yes
Pasteur	textual	speed	0.337	0.325	0.841	0.593	Yes
Grant-White	visual	textual	0.513	0.541	0.632	0.840	Yes
Grant-White	visual	speed	0.420	0.523	0.632	0.722	Yes
Grant-White	textual	speed	0.347	0.336	0.840	0.722	Yes

Note. Fornell–Larcker compares each factor's square root of AVE with the absolute fitted latent correlation. "Yes" means both square roots exceed that correlation; it is not a general validity verdict.